

11.7 Homework: Calculus Review

— Section A: No Calculator —

1. [Maximum mark: 4]

The function g is defined by $g(x) = e^{x^2+1}$, where $x \in \mathbb{R}$.

Find $g'(-1)$.

2. [Maximum mark: 5]

(a) The expression $\frac{3\sqrt{x} - 5}{\sqrt{x}}$ can be written as $3 - 5x^p$. Write down the value of p . [1]

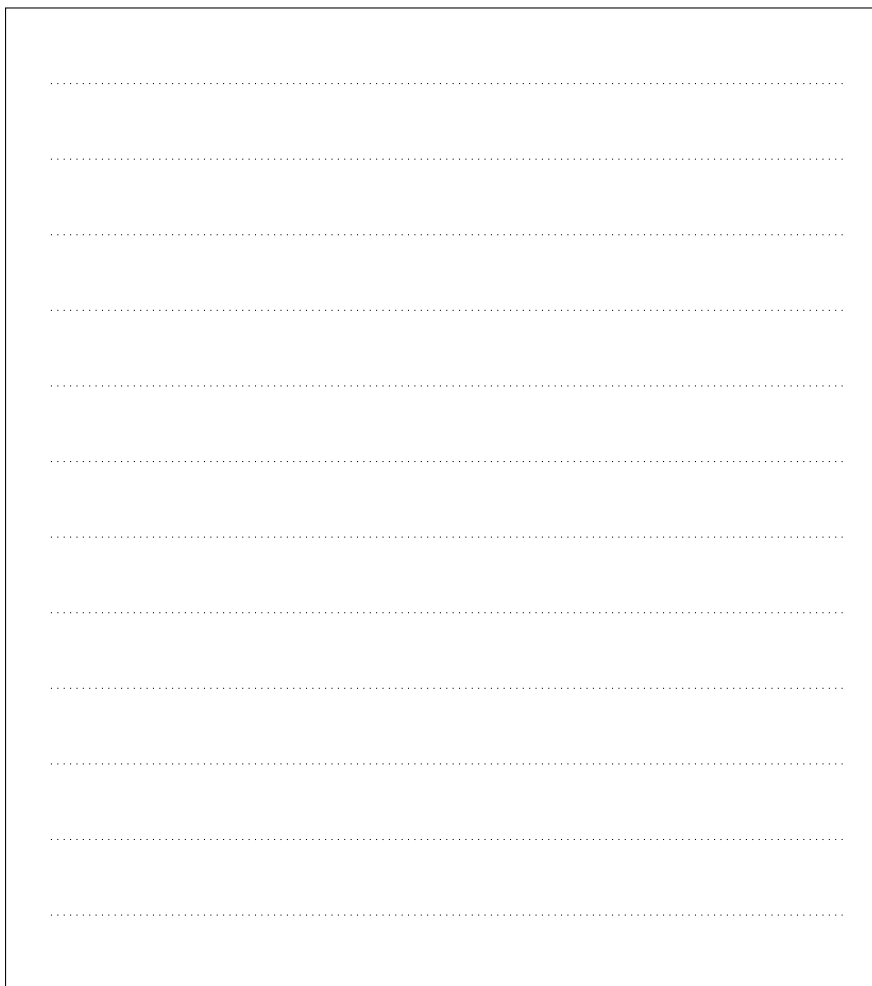
(b) Hence, find the value of $\int_1^9 \left(\frac{3\sqrt{x} - 5}{\sqrt{x}} \right) dx$. [4]

3. [Maximum mark: 5]

Consider the curve with equation $y = (2x - 1)e^{kx}$, where $x \in \mathbb{R}$ and $k \in \mathbb{Q}$.

The tangent to the curve at the point where $x = 1$ is parallel to the line $y = 5e^kx$.

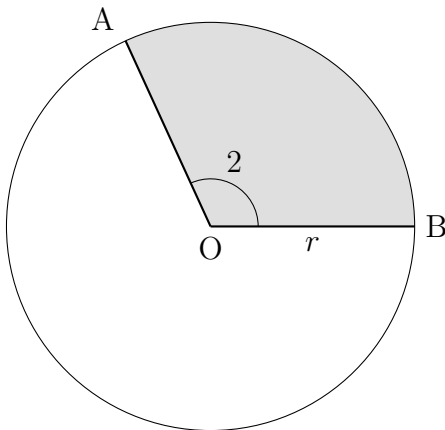
Find the value of k .



4. [Maximum mark: 6]

The following diagram shows a circle with centre O and radius r .

diagram not to scale



Points A and B lie on the circumference of the circle, and $\widehat{AOB} = 2$ radians.

The perimeter of the shaded region is 20.

(a) Find the value of r .

[3]

(b) Hence, find the exact area of the **non-shaded** region.

[3]

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5. [Maximum mark: 7]

The function f is defined for all $x \in \mathbb{R}$. The line with equation $y = 6x - 1$ is the tangent to the graph of f at $x = 4$.

(a) Write down the value of $f'(4)$. [1]

(b) Find $f(4)$. [1]

The function g is defined for all $x \in \mathbb{R}$ where $g(x) = x^2 - 3x$ and $h(x) = f(g(x))$.

(c) Find $h(4)$. [2]

(d) Hence find the equation of the tangent to the graph of h at $x = 4$. [3]

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— Section B: Calculator Allowed —

6. [Maximum mark: 6]

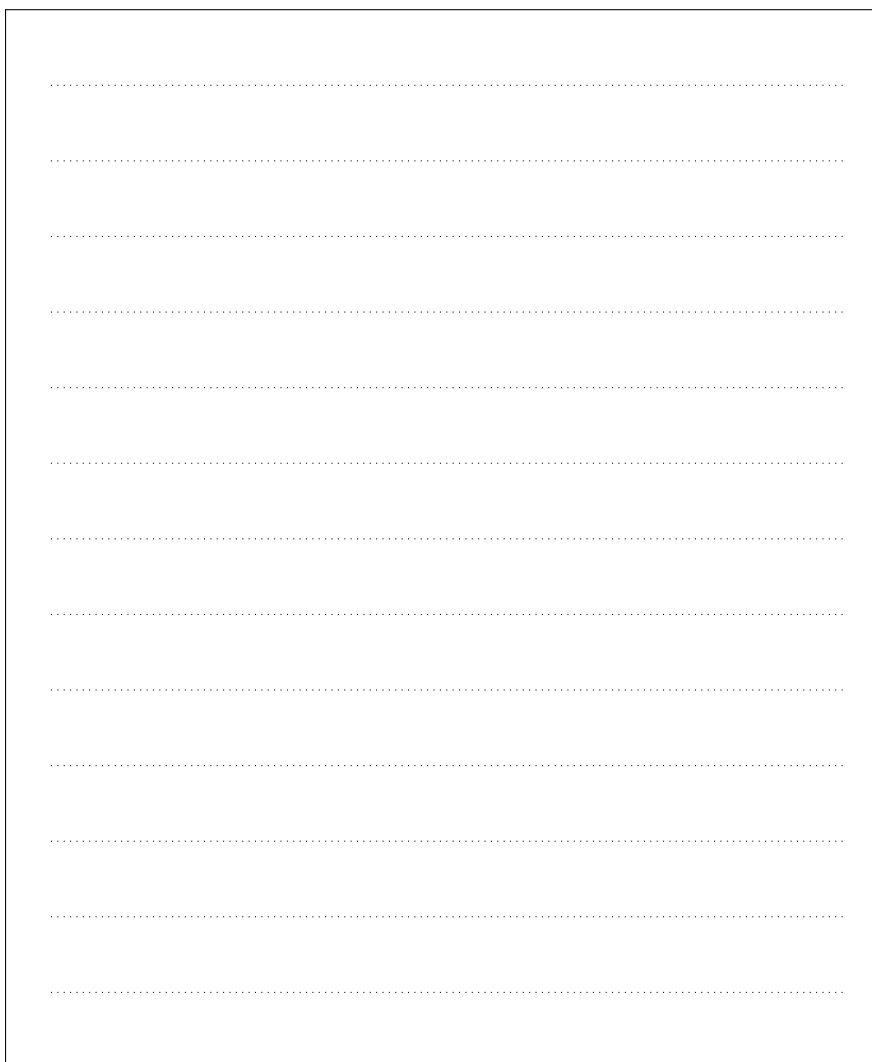
(a) Find $\int (6x + 7) \, dx$. [3]

(b) Given $f'(x) = 6x + 7$ and $f(1.2) = 7.32$, find $f(x)$. [3]

7. [Maximum mark: 5]

The derivative of a function g is given by $g'(x) = 3x^2 + 5e^x$, where $x \in \mathbb{R}$. The graph of g passes through the point $(0, 4)$.

Find $g(x)$.



8. [Maximum mark: 5]

The following table shows the daily high temperature (x °C) and the number of cold drinks sold (y) at a café on six days.

Temperature (x)	18	22	25	28	32	35
Drinks sold (y)	120	155	170	200	240	265

The regression line of y on x for this data can be written in the form $y = ax + b$.

- (a) Find the value of a and the value of b . [2]
- (b) Write down the value of the Pearson's product-moment correlation coefficient, r . [1]
- (c) Use the equation of your regression line to predict the number of cold drinks sold on a day when the high temperature is 30°C. Express your answer to the nearest integer. [2]

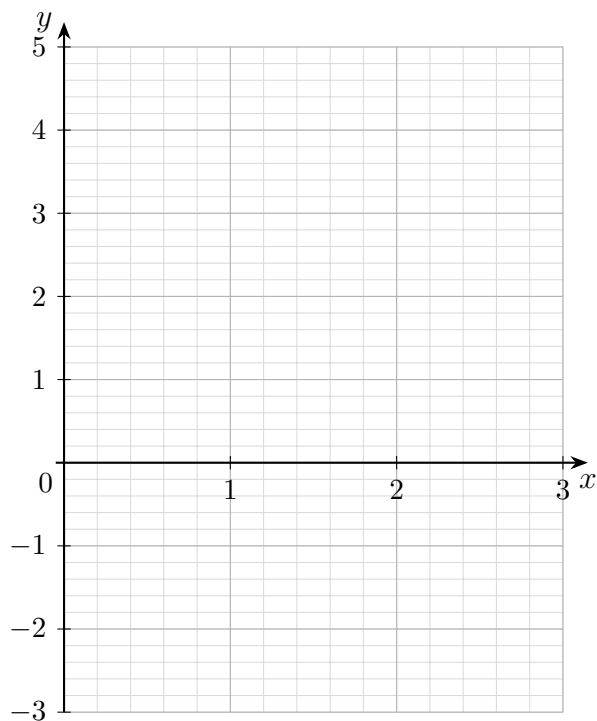
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9. [Maximum mark: 5]

Let $f(x) = 3x - 4^{0.15x^2}$ for $0 \leq x \leq 3$.

(a) Sketch the graph of f on the grid below.

[3]



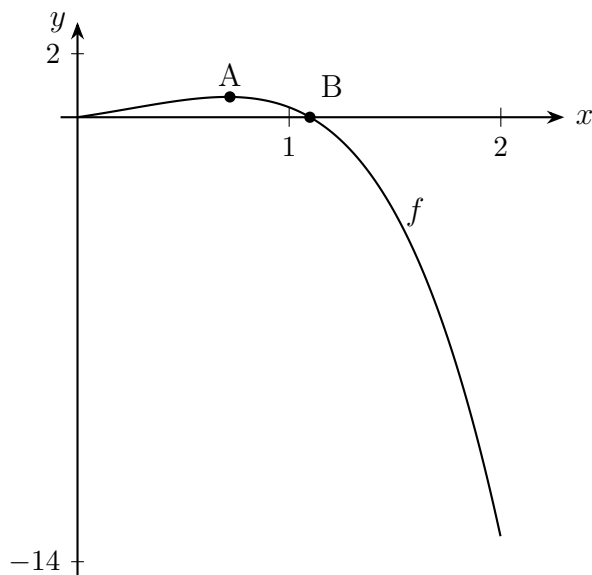
(b) Find the value of x for which $f'(x) = 0$.

[2]

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10. [Maximum mark: 6]

The function f is defined as $f(x) = \ln(xe^x + 1) - x^4$, for $0 \leq x \leq 2$. The graph of f is shown in the following diagram.



The graph of f has a local maximum at point A. The graph intersects the x -axis at the origin and at point B.

(a) Find the coordinates of A. [2]

(b) Find the x -coordinate of B. [1]

(c) Find the total area enclosed by the graph of f , the x -axis and the line $x = 2$. [3]

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